Multiple-Choice Questions on Recursion

1. Which of the following statements about recursion are true?

   I. Every recursive algorithm can be written iteratively.

   II. Tail recursion is always used in "divide-and-conquer" algorithms.

   III. In a recursive definition, a process is defined in terms of a simpler case of itself.

   (A) I only
   (B) I and II only
   (C) I and III only
   (D) II and III only

2. Which of the following, when used as the /* ***body*** */ of method sum, will enable that method to compute $1 + 2 + \ldots + n$ correctly for any $n > 0$?

```
/** Returns 1 + 2 + ... + n.
 *  Precondition: n is a positive
 * integer.
 */
public int sum(int n)
{
   /* body */
}
```

```
I. return n + sum(n - 1);
II. if (n == 1)
        return 1;
    else
        return n + sum(n - 1);
III. if (n == 1)
        return 1;
    else
        return sum(n) + sum(n - 1);
```

   (A) I only
   (B) II only
   (C) III only
   (D) I, II, and III

3. Refer to the method stringRecur.

```
public void stringRecur(String s)
{
    if (s.length() < 15)
        System.out.println(s);
    stringRecur(s + "*");
}
```

   When will method stringRecur terminate without error?

   (A) Only when the length of the input string is less than 15
   (B) Only when an empty string is input
   (C) For all string inputs
   (D) For no string inputs

4. Refer to method strRecur.

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Questions 5 and 6 refer to method `result`.

```
public int result(int n)
{
    if (n == 1)
        return 2;
    else
        return 2 * result(n - 1);
}
```

5. What value does `result(5)` return?

(A) 32
(B) 16
(C) 8
(D) 2

6. If $n > 0$, how many times will `result` be called to evaluate `result(n)` (including the initial call)?

(A) 2^n
(B) n
(C) $2n$
(D) n^2

7. Refer to method `mystery`.

```
public int mystery(int n, int a,
int d)
{
    if (n == 1)
        return a;
    else
        return d + mystery(n - 1,
        a, d);
}
```

What value is returned by the call `mystery(3, 2, 6)`?

(A) 20
(B) 14
(C) 8
(D) 2

8. Refer to method `f`.

```
public int f(int k, int n)
{
    if (n == k)
        return k;
    else
        if (n > k)
            return f(k, n - k);
        else
            return f(k - n, n);
}
```

What value is returned by the call `f(6, 8)`?

(A) 8
(B) 4
(C) 2
(D) 1

9. What does method `recur` do?

```
/** x is an array of n integers.
 * n is a positive integer.
 */
public int recur(int[] x, int n)
{
    int t;
    if (n == 1)
        return x[0];
    else
    {
        t = recur(x, n - 1);
        if (x[n-1] > t)
            return x[n-1];
        else
            return t;
    }
}
```

(A) It finds the largest value in x and leaves x unchanged.
(B) It finds the smallest value in x and leaves x unchanged.
(C) It sorts x in ascending order and returns the largest value in x.
(D) It sorts x in descending order and returns the smallest value in x.

10. Which best describes what the `printString` method below does?

```
public void printString(String s)
{
   if (s.length() > 0)
   {
      printString
      (s.substring(1));
      System.out.print
      (s.substring(0, 1));
   }
}
```

(A) It prints string s.
(B) It prints string s in reverse order.
(C) It prints only the first character of string s.
(D) It prints only the last character of string s.

11. Refer to the method power.

```
/** Returns base raised to the
 *  expo power.
 *  Precondition:
 *   - base is a nonzero real
 *     number.
 *   - expo is an integer.
 */
public double power(double base,
int expo)
{
   if (expo == 0)
      return 1;
   else if (expo > 0)
      return base * power(base,
      expo - 1);
   else
      return /* code */;
}
```

Which /* *code* */ correctly completes method power?
(Recall that $a^{-n} = 1/a^n$, $a \neq 0$; for example, $2^{-3} = 1/2^3 = 1/8$.)

(A) `(1 / base) * power(base, expo + 1)`
(B) `(1 / base) * power(base, expo - 1)`
(C) `base * power(base, expo + 1)`
(D) `base * power(base, expo - 1)`

12. Consider the following method.

```
public void doSomething(int n)
{
   if (n > 0)
   {
      doSomething(n - 1);
      System.out.print(n);
      doSomething(n - 1);
   }
}
```

What would be output following the call `doSomething(3)`?

(A) `1121213`
(B) `1213121`
(C) `1211213`
(D) `1123211`

13. A user enters several positive integers at the keyboard and terminates the list with a sentinel (-999). A writeEven method reads those integers and outputs the even integers only, in the reverse order that they are read. Thus, if the user enters

```
3 5 14 6 1 8 -999
```

the output for the writeEven method will be

```
8 6 14
```

Assume that the user enters at least one positive integer and terminates the list with −999. Here is the method.

```
/** Postcondition: All even
 *  integers in the list are
 *  output in reverse order.
 */
public static void writeEven()
{
    int n = ...; //read user input
    if (n != -999)
    {
        /* code */
    }
}
```

Which /* *code* */ satisfies the postcondition of method writeEven?

I.
```
if (n % 2 == 0)
    System.out.print(n + " ");
writeEven();
```

II.
```
if (n % 2 == 0)
    writeEven();
System.out.print(n + " ");
```

III.
```
writeEven();
if (n % 2 == 0)
    System.out.print(n + " ");
```

(A) II only
(B) III only
(C) I and II only
(D) I, II, and III

14. Refer to the following recursive method.

```
public int mystery(int n)
{
    if (n < 0)
        return 2;
    else
        return mystery(n - 1)
               + mystery(n - 3);
}
```

What value is returned by the call mystery(3)?

(A) 12
(B) 10
(C) 8
(D) 6

Questions 15 and 16 refer to method `t`.

```
/** Precondition: n is a positive
    integer. */
public int t(int n)
{
    if (n == 1 || n == 2)
        return 2 * n;
    else
        return t(n - 1) - t(n - 2);
}
```

15. What will be returned by `t(5)`?

(A) 4
(B) 2
(C) −2
(D) −4

16. For the method call `t(6)`, how many calls to `t` will be made, including the original call?

(A) 6
(B) 11
(C) 15
(D) 25

17. This question refers to methods `f1` and `f2` that are in the same class.

```
public int f1(int a, int b)
{
    if (a == b)
        return b;
    else
        return a + f2(a - 1, b);
}

public int f2(int p, int q)
{
    if (p < q)
        return p + q;
    else
        return p + f1(p - 2, q);
}
```

What value will be returned by a call to `f1(5, 3)`?

(A) 5
(B) 7
(C) 12
(D) 15

18. Consider method `foo`.

```
public int foo(int x)
{
    if (x == 1 || x == 3)
        return x;
    else
        return x * foo(x - 1);
}
```

Assuming no possibility of integer overflow, what will be the value of z after execution of the following statement? Note that $n! = (n)(n-1)(n-2)\ldots(2)(1)$.

```
int z = foo(foo(3) + foo(4));
```

(A) (15!)/(2!)
(B) 3! + 4!
(C) (7!)!
(D) (3! + 4!)!

Questions 19 and 20 refer to the `IntFormatter` class below.

```
public class IntFormatter
{
   /** Write 3 digits adjacent to each other.
    *  Precondition: n is a nonnegative integer.
    */
   public static void writeThreeDigits(int n)
   {
      System.out.print(n / 100);
      System.out.print((n / 10) % 10);
      System.out.print(n % 10);
   }
   /** Insert commas in n, every 3 digits starting at the right.
    *  Precondition: n is a nonnegative integer.
    */
   public static void writeWithCommas(int n)
   {
      if (n < 1000)
         System.out.print(n);
      else
      {
         writeThreeDigits(n % 1000);
         System.out.print(",");
         writeWithCommas(n / 1000);
      }
   }
}
```

19. The method `writeWithCommas` is supposed to print its nonnegative `int` argument with commas properly inserted (every three digits, starting at the right). For example, the integer 27048621 should be printed as 27,048,621. Method `writeWithCommas` does not always work as intended, however. Assuming no integer overflow, which of the following integer arguments will not be printed correctly?

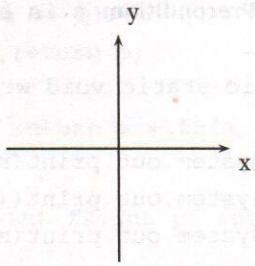
(A) 896
(B) 251462251
(C) 365051
(D) 4

20. Which change in the code of the given methods will cause method `writeWithCommas` to work as intended?

(A) Interchange the lines `System.out.print(n / 100)` and `System.out.print(n % 10)` in method `writeThreeDigits`.
(B) Interchange the lines `writeThreeDigits(n % 1000)` and `writeWithCommas(n / 1000)` in method `writeWithCommas`.
(C) In the method `writeWithCommas`, change the line `writeThreeDigits(n % 1000)` to `writeThreeDigits(n / 1000)`.
(D) In the method `writeWithCommas`, change the recursive call `writeWithCommas(n / 1000)` to `writeWithCommas(n % 1000)`.

21. Consider the following method.

```
public static void sketch(int x1, int y1, int x2, int y2, int n)
{
    if (n <= 0)
        drawLine(x1, y1, x2, y2);
    else
    {
        int xm = (x1 + x2 + y1 - y2) / 2;
        int ym = (y1 + y2 + x2 - x1) / 2;
        sketch(x1, y1, xm, ym, n - 1);
        sketch(xm, ym, x2, y2, n - 1);
    }
}
```

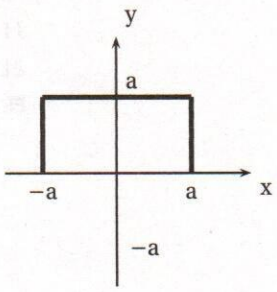


Assume that the screen looks like a Cartesian coordinate system with the origin at the center, and that `drawLine` connects (`x1`,`y1`) to (`x2`,`y2`). Assume also that `x1`, `y1`, `x2`, and `y2` are never too large or too small to cause errors. Which picture best represents the sketch drawn by the method call

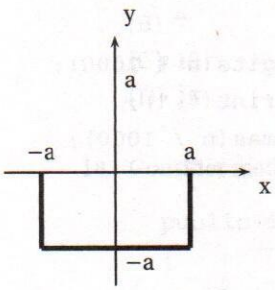
```
sketch(a, 0, -a, 0, 2)
```

where a is a positive integer?

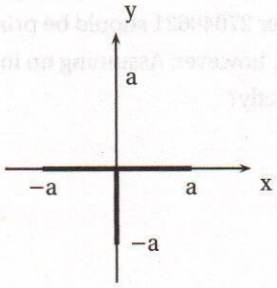
(A)



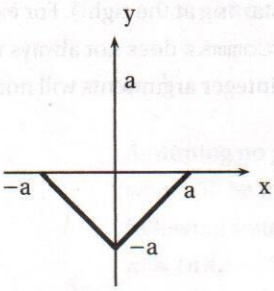
(B)



(C)



(D)



22. Which of the following is a valid reason why merge sort is a better sorting algorithm than insertion sort for sorting long, randomly ordered lists?

I. Merge sort requires less code than insertion sort.

II. Merge sort requires less storage space than insertion sort.

III. Merge sort runs faster than insertion sort.

(A) I only
(B) III only
(C) I and II only
(D) II and III only

23. Consider the following method.

```
/** Precondition: a[0],a[1]...a[n-1]
 *  contain integers. */
public static int someMethod(
        int[] a, int n, int value)
{
    if (n == 0)
        return -1;
    else
    {
        if (a[n-1] == value)
            return n - 1;
        else
            return someMethod
            (a, n - 1, value);
    }
}
```

The method shown is an example of

(A) merge sort.
(B) selection sort.
(C) binary search.
(D) sequential search.

Optional Topic

24. The `partition` method for quicksort partitions a list as follows.

I. A pivot element is selected from the array.

II. The elements of the list are rearranged such that all elements to the left of the pivot are less than or equal to it; all elements to the right of the pivot are greater than or equal to it.

Partitioning the array requires which of the following?

(A) A recursive algorithm
(B) A temporary array
(C) A swap algorithm for interchanging array elements
(D) A merge method for merging two sorted lists

25. Assume that merge sort will be used to sort an array `arr` of `n` integers into increasing order. What is the purpose of the `merge` method in the merge sort algorithm?

(A) Partition `arr` into two parts of roughly equal length, then merge these parts.
(B) Use a recursive algorithm to sort `arr` into increasing order.
(C) Divide `arr` into `n` subarrays, each with one element.
(D) Merge two sorted parts of `arr` into a single sorted array.